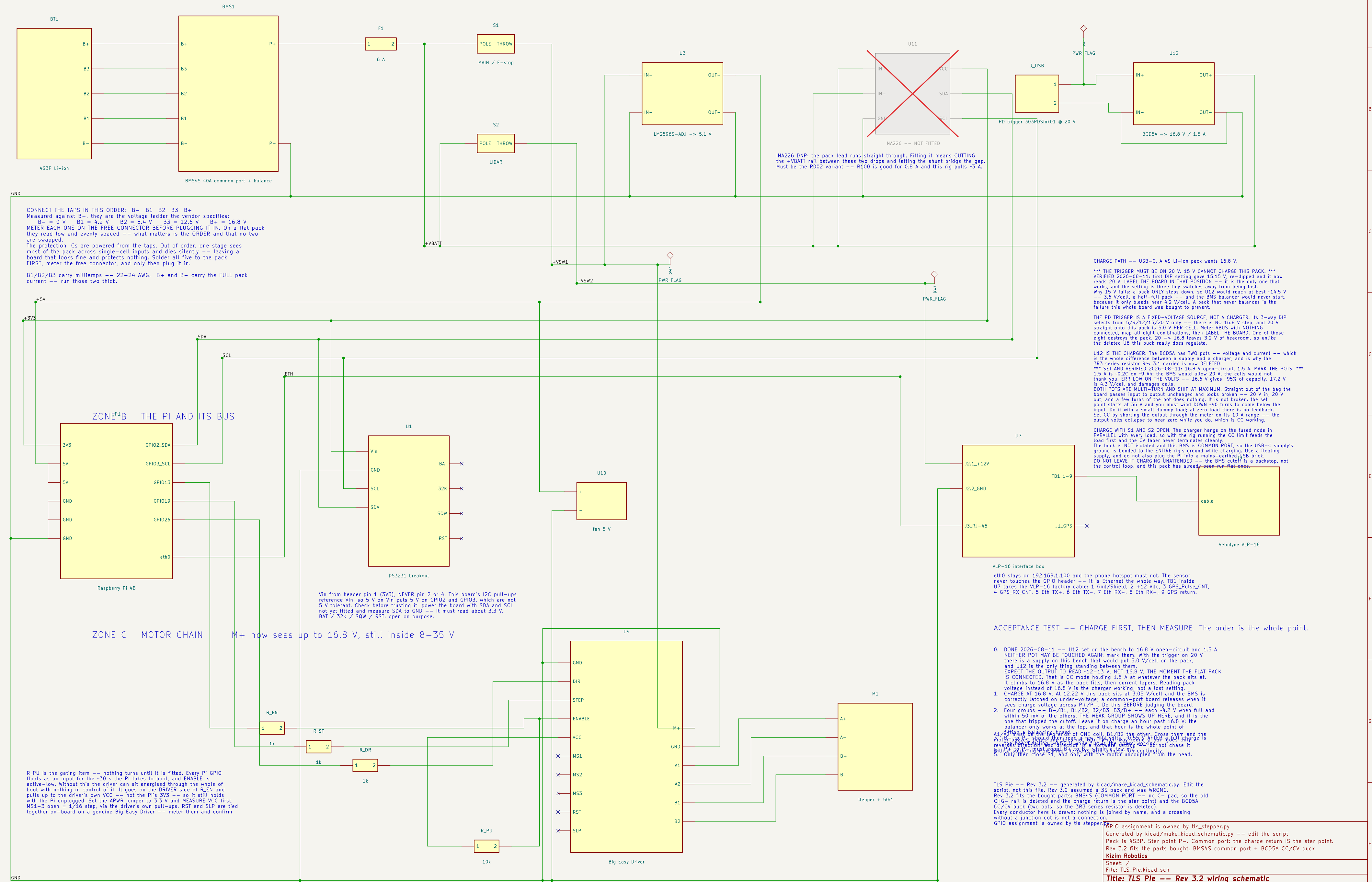


ZONE A POWER CHAIN 4S3P pack, common-port balancing BMS, USB-C charge

S1 and S2 both hang off the fused node, in PARALLEL. S1 does NOT switch the lidar -- opening S1 kills the Pi and the motor and leaves the VLP-16 spinning on S2.

*** NEW ON 4S: S2 HANDS THE VLP-16 UP TO 16.8 V ***
On the old 3S assumption that leg never passed 12.6 V. CHECK THE SENSOR'S OWN INPUT RANGE before S2 is ever closed. If it will not take 16.8 V that leg needs a regulator -- the one place going to 4S makes things worse.



ACCEPTANCE TEST -- CHARGE FIRST, THEN MEASURE. The order is the whole point.

- [illegible]

TIS Pie -- Rev 3.2 -- generated by kicad/make_kicad_schematic.py. Edit the script, not this file. Rev 3.0 assumed a 35 pack and was WRONG.
 Rev 3.2 fits the bought parts: BM545 (COMMON PORT --- no C- pad, so the o CHG- rail is deleted and the charge return is the star point) and the BCD5A CC/CV buck (two pots, so the 3R3 series resistor is deleted).
 Every conductor here is drawn: nothing is joined by name, and a crossing without a junction dot is not a connection.
 GPIO assignment is owned by tIs-stepperPIGPIO assignment is owned by tIs-

P2PID assignment is owned by TLS_stepper.py
 Generated by kicad/make_kicad_schematic.py -- edit the script
 Pack is 453P. Star point P--. Common port: the charge return IS the star point.
 Rev 3.2 fits the parts bought: BMS45 common port + BCD54 CC/CV buck
Kizim Robotics
 Sheet: /
 File: TLS_Pie_Kicad_sch

Title: TLS Pie --- Rev 3.2 wiring schematic

Size: A2	Date: 2026-08-11	Rev: 3.
KiCad E2.A. 10.0.5		Id: 1/1